Xianliang Li

PhD student at the Institute of Statistical Mathematics
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Tokyo, Japan

Research Interests

Optimizer, Foundation Model Training, Deep Learning Theory.

Education

2025 – 2028 (Expected) **The Graduate University for Advanced Studies, SOKENDAI**
Ph.D. in Statistical Science (Supervisor: Prof. Han Bao)

2022 – 2025 **University of Chinese Academy of Sciences**
M.Eng. in Computer Technology (Supervisor: Prof. Sheng Xu)

2018 – 2022 **Sun Yat-sen University**
B.Sc. in Physics

Research Experience

October 2025 – Present **The Institute of Statistical Mathematics**
Research Assistant (Host: Prof. Han Bao)

July 2025 – Present **The Chinese University of Hong Kong**
Visiting Student (Host: Prof. Weiyang Liu)

September 2023 – June 2025 **Shenzhen Institutes of Advanced Technology**
Research Student (Host: Prof. Sheng Xu)

Honors and Scholarships

2025 – Present **SOKENDAI Special Researcher** (The Graduate University for Advanced Studies, SOKENDAI)
Supported by JST SPRING, Japan Grant Number JPMJSP2104.

Publications

** - equal contribution, † - corresponding author.*

2026 **Optimal Sensor Placement Using Decentralized TDOA Sensor Network in Source Localization**
Kang Tang, Sheng Xu†, **Xianliang Li**, K. C. Ho, He Kong, Kutluyil Dogancay.
IEEE Transactions on Aerospace and Electronic Systems.

2025 **3D AOA Target Tracking Using Multiple Distributed UAVs with Unknown Self-Localizations and Dynamic Anchors**
Rongrong Xu, Sheng Xu†, Xianyong Mu, **Xianliang Li**, Dashuai Wang, Tiantian Xu.
IEEE International Conference on Real-time Computing and Robotics.
[IEEE Xplore]

2025 **Time-of-Arrival Simultaneous Sensor and Target Localization with Dynamic Optimal Sensor Placements**
Xianliang Li, Sheng Xu†, K. C. Ho.
IEEE Transactions on Aerospace and Electronic Systems.
[IEEE Xplore]

2025 **AOA Sensor Placement for Anchor-Assisted Target Localization in GNSS-Denied Environment: Formulation, Bounds and Optimization**
Sheng Xu, Linlong Wu, **Xianliang Li**, Xinyu Wu†, Tiantian Xu†.
IEEE Transactions on Mobile Computing.
[IEEE Xplore]

2025 **On the Performance Analysis of Momentum Method: A Frequency Domain Perspective**
Xianliang Li*, Jun Luo*, Zhiwei Zheng*, Hanxiao Wang, Li Luo, Lingkun Wen, Linlong Wu, Sheng Xu†.
International Conference on Learning Representations (ICLR).
[OpenReview]

2024 **Systematical Sensor Path Optimization Solutions for AOA Target Localization Accuracy Improvement with Theoretical Analysis**
Sheng Xu, Bing Zhu, **Xianliang Li**, Xinyu Wu, Tiantian Xu†.
IEEE Transactions on Vehicular Technology.
[IEEE Xplore]

2024 **3D Source Tracking Using a Position-Unknown AOA Sensor with Measurement Drift and UAV Moving Direction Optimization**
Rongrong Xu, Sheng Xu[†], Kaimin Cao, **Xianliang Li**, Jialiang Wang, Wujing Cao.
IEEE International Conference on Real-time Computing and Robotics.
[IEEE Xplore]

Professional Activities

Journal Reviewer

Digital Signal Processing (DSP).

Conference Reviewer

NeurIPS (2026), ICLR (2025).